Run report: étale-cover search for T_n (lead L8)

run-20261018T2043Z · study `research/ecdlp_leads_20261018` · commit `bd694bf2`

Kind. Exact enumeration with the native Rust crate `etale_search`. Not a timing or statistical measurement. Host: shared cloud container (Xeon 2.80 GHz, rustc 1.97.0), isolation L0.

What ran. Every smooth curve $B : y^2 + h(x)y = f(x)$ over $\mathbb{F}_2$ of genus 2, and every smooth hyperelliptic curve of genus 3. For each, $J_B(\mathbb{F}_2)$ was computed from relations among places of degree at most 4, validated by free rank 1 and torsion $= L_B(1)$. Then, for every odd prime $n \mid \#J_B(\mathbb{F}_2)$ and every character χ of $J_B(\mathbb{F}_2)/n$, $L(B, \chi, X)$ was built and tested for the factor $(1 - c\pi X)(1 - c\overline{\pi}X)$, with $c \in \mu_n$ and $\pi = \left(t \pm i\sqrt{7}\right)/2$, $t = \pm 1$.

quantity	genus 2	genus 3 (hyperelliptic)
smooth models	768	6,144
validated, with an odd prime dividing $\#J$	520	4,128
failed validation	0	0
failed the RH check	0	0
largest $\#J_B(\mathbb{F}_2)$ among them	19	56
characters with a T_n factor	192, all $n = 3$	576, all $n = 3$

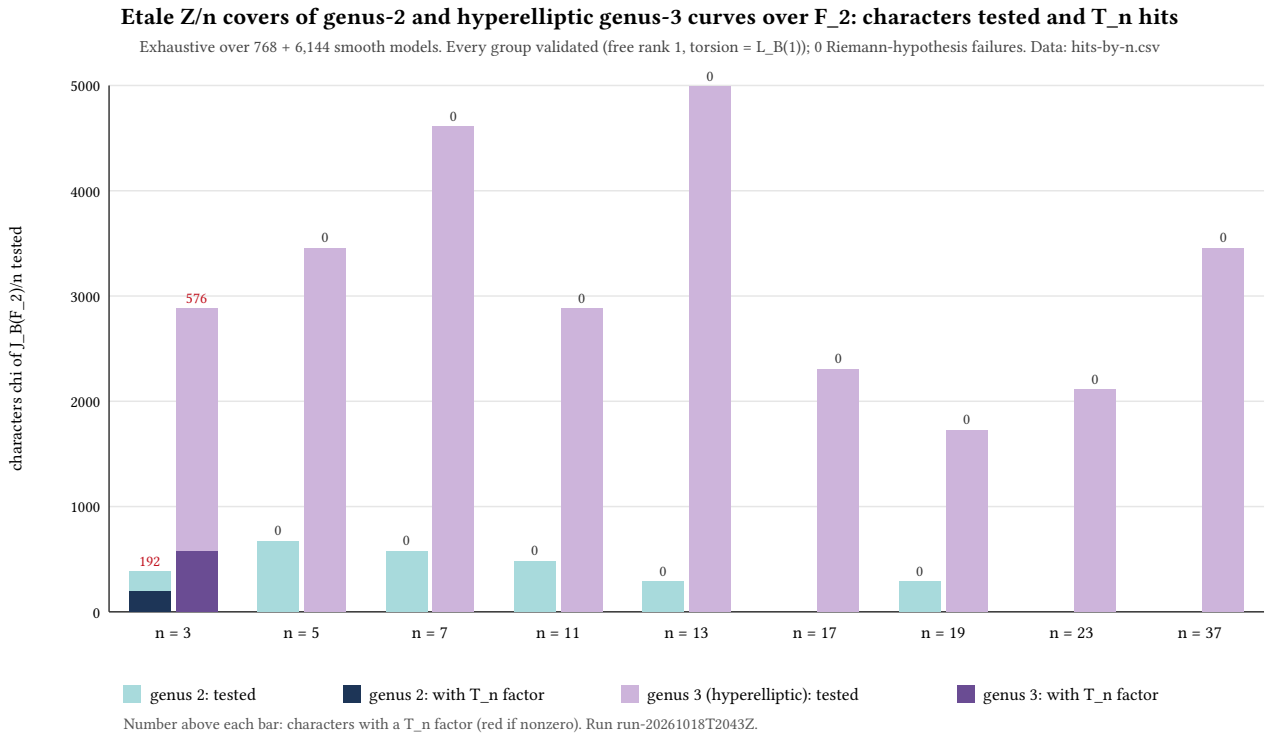


Figure 1: Characters tested and characters with a T_n factor, per n. Recorded values from `hits-by-n.csv`.

Hits occur in six isogeny classes of B, three in genus 2 and three in genus 3 (`hit-isogeny-classes.csv`). They cover both $a = 1$ and $a = 0$. No character has a T_n factor for any $n \geq 5$.

Not covered. Plane-quartic genus-3 curves, which are needed for $\#J \geq 57$ and so for $n = 131, 163$. Genus 4 and above. Ramified covers.

Records. `RUN.txt` (commit, host, SHA-256 of outputs), `genus2.jsonl`, `genus3.jsonl`, `hits-by-n.csv`, `hit-isogeny-classes.csv`, `hits-by-n.plot.awk`.